CS5260 Neural Networks and Deep Learning II
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Project Proposal

Group 9

Student Name	Student Number	Email
Ang Wei Hao Desmond	A0093896H	e0011769@u.nus.edu
Koh Kai Yit	A0200100U	e0407081@u.nus.edu
Lakshminarayanan Kaushik Rangaraj	A0214270U	e0518554@u.nus.edu
See Cheng Jiang	A0218252L	e0544288@u.nus.edu

Motivation

The world is a huge place, with endless places to explore and experience. Working adults have limited time to plan and implement their travel itinerary to enjoy the limited amount of vacation time that they have. Travel planning is inherently complex, ambiguous, and comprises many different tasks, such as finding flights, curating activities, managing budgets, requiring multiple specialised skill sets. Hence, we would like to leverage the power of AI to automate the travel logistics process and provide a seamless travel experience for our users.

Objectives

The objectives of the project are:

- Design a multi-agent AI system capable of decomposing complex travel prompts into structured planning tasks.
- Implement domain-specific agents to handle transport, accommodation and experience/activities planning.
- Incorporate a shared global constraint management mechanism to ensure consistency across agents.
- Implement a verification mechanism to detect and resolve logical conflicts.
- Support multimodal input such as images and website links.
- Deploy the system as a publicly accessible web application.
- Ensure inference cost remains within the SGD 10 testing constraint.

Travel Planner Agent Architecture

An Ingestion Agent will first process multimodal inputs such as images and website links into structured text representations that can be understood by the coordinator. The processed input will then be passed into a hierarchical multi-agent system consisting of three levels.

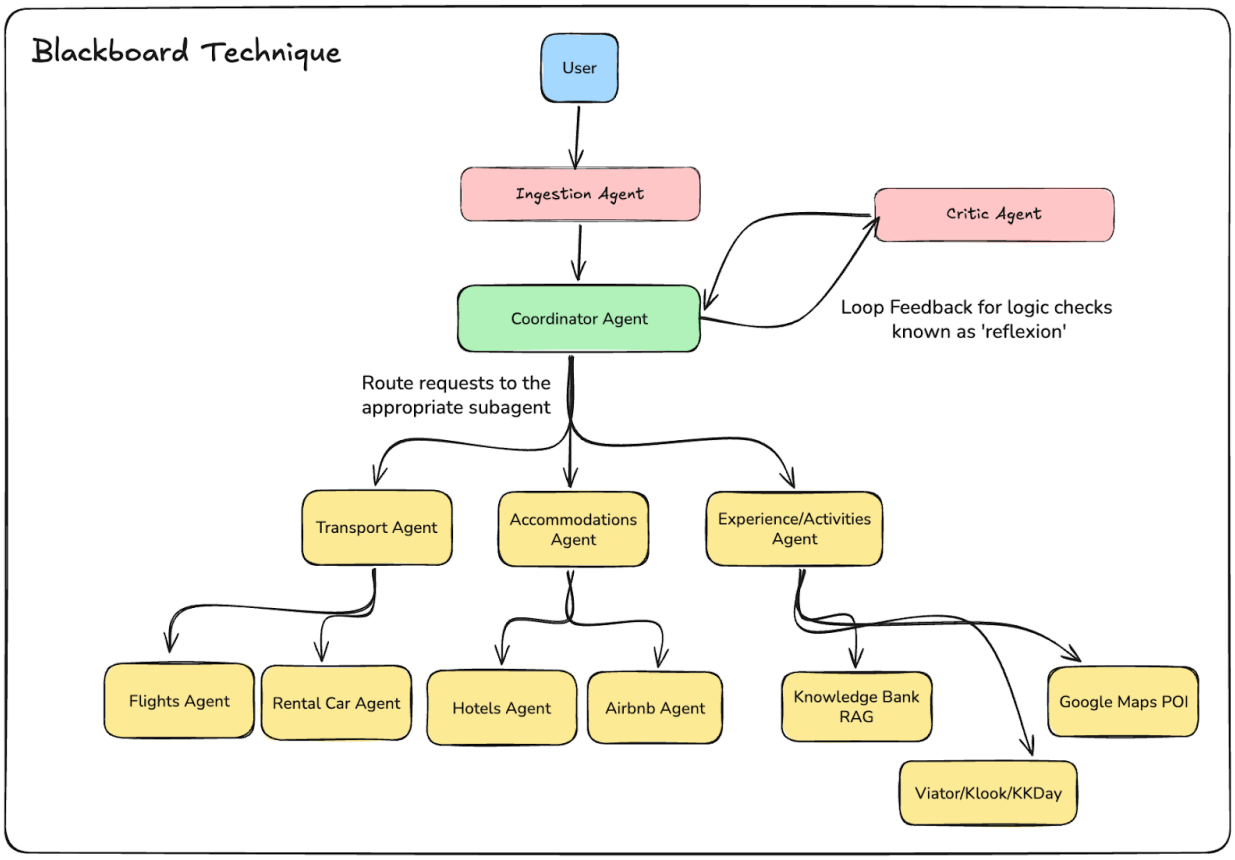
Level 1: Root Coordinator Agent
The Root Coordinator will receive the processed user input and perform high-level task decomposition. It will determine the scope of the request and delegate sub-tasks to specialised domain agents.

Level 2: Domain Agents
Domain-specific agents such as Transport, Accommodation, Experiences/Activities agents will manage planning within their respective domains. They will interpret constraints such as budget, time and preferences, and further decompose tasks into executable actions.

Level 3: Worker Agents
Worker agents will perform concrete operations such as search, retrieval and structured data generation. These agents will interact with external tools such as web search APIs to gather information required for itinerary construction.

To prevent conflict when agents work in parallel under shared constraints such as budget, a shared global state (Blackboard mechanism) will be used to allow agents to read and update constraints dynamically, ensuring coordination across parallel tasks.

A Critic Agent will then perform final consistency checks before the itinerary is presented to the user.



Evaluation Metrics

Quantitative Metrics	Qualitative Metrics
Success Rate: Percentage of generated itineraries that have no logical conflicts **Latency**: Average time to generate a full itinerary **Cost per Query**: To ensure we stay within the SGD 10 limit	**User Satisfaction**: 1-5 star rating on the final itinerary

Limitations

We note that the agents may hallucinate and we're unable to test the agent functions on live servers or live bookings. Hence, we propose for agents to output strict JSON formats by using Pydantic models. Furthermore, most of the knowledge would be done via google web-search and this might result in poorer performance for travelling in China due to the Great Firewall.